

ParkTrack

Self-monitoring of Parkinson's Symptoms to Support Clinical Research

Bachelor Thesis



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August, 2025

By

Mozhdah Atai

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Signature

..... August 1, 2025

Date

Abstract

Parkinson's Disease (PD) research requires access to standardized, high-quality, and patient-generated data on motor and non-motor symptoms. This project presents ParkTrack, a conceptual design for a mobile platform integrated with wearable devices to support self-monitoring of PD symptoms, including tremor, bradykinesia, sleep, heart rate, and medication intake. Using an iterative, user-centered design process, the prototype was co-designed with researchers, patients, and healthcare professionals to create a usable and research-oriented solution. ParkTrack provides automated data collection through wearables, manual input options, and visual feedback, aiming to bridge the gap between patient self-monitoring and clinical research needs.

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1 Introduction

Parkinson's Disease (PD) is a chronic, progressive neurodegenerative disorder that affects more than 10 million people worldwide, with approximately 12,000 individuals diagnosed in Denmark [1]. It results from the degeneration of dopamine-producing neurons in the substantia nigra and leads to a wide spectrum of motor and non-motor symptoms. These symptoms vary significantly between individuals and evolve over time, making PD particularly challenging to monitor and manage in both clinical and research contexts [2, 3].

Traditional PD assessments are typically conducted during brief clinical consultations, providing only limited snapshots of a patient's condition. These evaluations fail to capture daily symptom fluctuations, treatment responses, and overall well-being, leaving clinicians and researchers without the continuous, real-world data needed to fully understand disease progression [4].

Mobile health (mHealth) refers to the use of mobile devices, applications, and connected technologies to support health monitoring, data collection, and patient engagement. In the context of PD, mHealth and wearable sensor technologies offer promising opportunities to address these gaps by providing continuous monitoring and structured self-reporting, improving data collection for both patient care and research [5].

Research on wearable systems for PD highlights their potential to monitor key symptoms such as tremor, bradykinesia (slow movements), and gait disturbances (e.g., falls, freezing of gait, balance issues) in real-life settings [6]. In addition, studies show that digital tools can actively involve patients in managing their condition while providing structured, high-quality data that can be valuable for research and also contribute to patient empowerment by increasing their involvement and sense of control over disease management [7].

Beyond its research potential, such data collection can also support clinical practice by providing healthcare professionals (HCP) with richer insights into patients' daily symptom fluctuations, treatment responses, and overall well-being. This can enhance clinical decision-making, improve patient-clinician communication, and ultimately help tailor treatment plans more effectively to individual patients' needs.

Despite these advances, current mobile applications are mainly designed for symptom tracking in self-care or clinical contexts and often lack the comprehensive, long-term data collection needed for research [4]. This creates challenges for researchers in obtaining standardized, multimodal datasets that reflect patients' lived experiences.

This bachelor's thesis addresses this gap by designing a mobile application, ParkTrack, to enable early- to mid-stage PD patients to self-monitor both motor and non-motor symptoms in a structured, research-oriented manner.

1.1 Research Questions

To achieve these objectives, this thesis is guided by the following research questions:

How can a mobile application be designed to enable early- to mid-stage PD patients to self-monitor motor and non-motor symptoms in a structured way that supports research data collection?

To explore this, the following sub-questions are investigated:

1. How can co-creation principles be applied to design a mobile application that is both usable and acceptable for early- to mid-stage PD patients?
2. What types of symptom and health data are most relevant for researchers, and how can these be effectively captured and visualized?
3. How can patient engagement and adherence to self-monitoring be supported through design?

1.2 Goals and Methods

Based on these challenges, this thesis aims to design a solution that addresses the identified gaps through the following goals and methods: The primary goal of this thesis is to design and prototype a mobile application, ParkTrack, that supports patient-driven self-monitoring of motor and non-motor symptoms and delivers structured datasets for research purposes. The system is designed to fulfill the following goals:

Enable Multi-Modal Symptom Tracking: Combine wearable sensor data with patient-reported outcomes to capture both motor (e.g., tremor, bradykinesia) and non-motor (e.g., sleep, cognitive performance) symptoms.

Support Medication Management: Provide tools for logging medication intake and assessing its perceived effectiveness over time, enabling researchers to study medication adherence and response patterns.

Facilitate Research-Oriented Data Collection: Structure data to facilitate its usability for clinical research, with a focus on longitudinal tracking and integration of multiple symptom domains.

Improve Usability and Accessibility: Design the prototype to be easy to navigate and accessible for users with PD, minimizing barriers such as login difficulties and ensuring an intuitive interface for daily use.

Promote Patient Engagement: Design features that encourage consistent use through intuitive interfaces and meaningful feedback, ensuring sustained participation in self-monitoring activities.

The project employs an iterative, user-centered design (UCD) methodology as outlined in The UX Book [8]. The design process of ParkTrack followed the UX lifecycle proposed by Hartson and Pyla, as illustrated in Figure 3.1. This ensured a structured and iterative

approach, where continuous involvement of patients, caregivers, and healthcare professionals informed the design through multiple feedback loops.

This includes user needs analysis, wireframing, and prototyping using Figma, followed by formative feedback sessions with a small group of patients, relatives, and healthcare professionals. Data for prototype design are simulated based on literature and existing research on wearable technologies. This approach aims to address real-world research needs while striving to remain feasible and acceptable for its intended users.

2 Background and Related Work

Parkinson's Disease (PD) is a chronic and progressive neurodegenerative disorder that affects more than 10 million people worldwide, with prevalence increasing with age [9]. It results from the degeneration of dopamine-producing neurons in the substantia nigra, leading to a wide range of motor and non-motor symptoms. The degeneration of these neurons and its impact on motor control is illustrated in Figure 2.1.

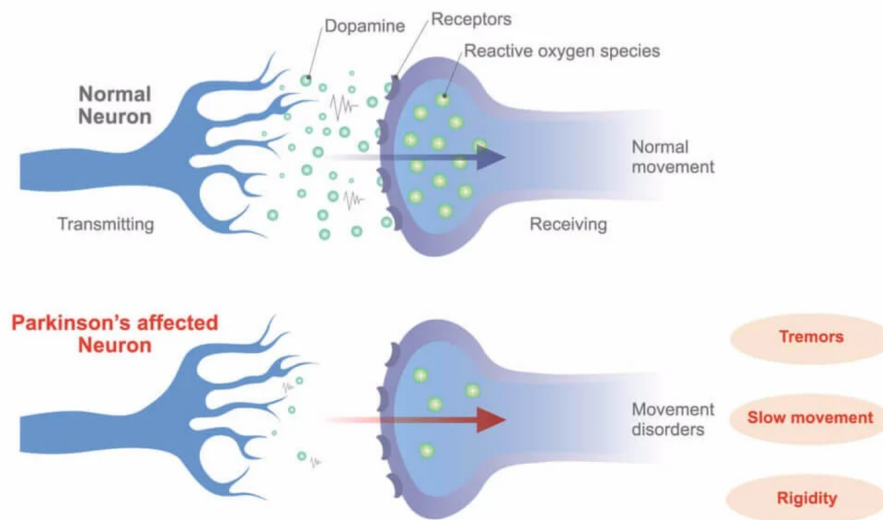


Figure 2.1: The changes that occur in the neurons of patients with PD. Taken from [10].

Motor symptoms include tremor, bradykinesia, rigidity, and postural instability, which are typically visible and interfere with daily activities, whereas non-motor symptoms such as cognitive decline, mood disturbances, hallucinations, autonomic dysfunction, and sleep disorders are less visible but significantly impact quality of life [11, 3].

The progression of PD is commonly classified using the Hoehn and Yahr scale (H&Y), which divides the disease into five stages [12]. In the early to mid-stages (stages 1–3), patients often retain enough independence to engage in self-monitoring and use digital health tools, while advanced stages (4–5) involve more severe physical and cognitive impairments, limiting their ability to participate in independent symptom tracking. However, every disease course in terms of progression time and severity is individual and does not follow a single-way path.

Symptom severity and presentation vary widely between individuals and can fluctuate considerably throughout the day, posing challenges for clinical management. Traditional assessments, typically performed during brief clinical consultations, offer only a limited snapshot of the patient's condition and fail to capture symptom variability, treatment responses, and daily fluctuations [13]. These limitations highlight the need for continuous and objective symptom monitoring to improve both patient care and research outcomes.

2.1 Clinical Assessment Approaches

The clinical evaluation of PD symptoms often relies on standardized, clinical rating scales such as the Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) and the H&Y staging system [6, 12].

These tools require in-person consultations and are usually administered periodically, which limits their ability to capture day-to-day symptom variations. Moreover, such assessments depend heavily on subjective reporting and clinical observation, which may not fully reflect the complexity of motor and non-motor symptom dynamics.

2.2 Digital Health and Smartwatch Technologies

To address the limitations of episodic assessments, mHealth and smartwatch-based monitoring have gained increasing attention in recent years. These solutions enable continuous and objective monitoring of PD symptoms in daily life, offering detailed insights into motor fluctuations and treatment responses [14].

Smartwatches equipped with inertial measurement units (IMU), such as the Apple Watch, can passively capture tremor, bradykinesia, dyskinesia, gait parameters, and postural changes.

A recent review by Moreau et al. summarized the most commonly used monitoring systems in Europe, such as PDMonitor, Parkinson KinetiGraph (PKG), and STAT-ON, emphasizing their potential to improve treatment management and research data collection [6]. In addition, Apple has introduced the CMMovementDisorderManager API, which allows developers to collect tremor and dyskinesia data from Apple Watch users. This API enables passive, power-efficient monitoring of symptoms while ensuring compliance with HealthKit privacy guidelines and regulatory requirements [15].

2.3 Existing Solutions and Prior Work

Several studies have demonstrated the potential of integrating self-tracking technologies into PD management. Kuosmanen explored the role of technological support for self-care, highlighting how self-monitoring can empower patients and contribute valuable data for research [5]. Similarly, Channa et al. presented a cloud-connected bracelet that uses wearable sensors and machine learning algorithms to continuously monitor motor and non-motor symptoms, showing promising results for remote data collection in research contexts [4].

In addition, Moreau et al. provided a comprehensive review of body-worn monitoring devices for PD management (Figure 2.2). Among these, the PKG stands out for its single-wrist sensor design, which records bradykinesia, dyskinesia, and tremor over 6–7 days with a home-based docking system that allows patients to log into a portal to review and compare their data on a weekly basis.

Studies have shown that PKG data can meaningfully inform clinical decisions and reduce motor complications when integrated into treatment planning. The STAT-ON system,

worn at the waist, offers real-time gait analysis and fluctuation detection using validated machine learning algorithms, although its accuracy decreases for mild or isolated limb dyskinesias.

PDMonitor, in contrast, employs a multi-sensor setup across five body locations to provide detailed reports on motor states, gait, and freezing of gait episodes, but its complex setup and report interpretation may limit patient usability in everyday life.

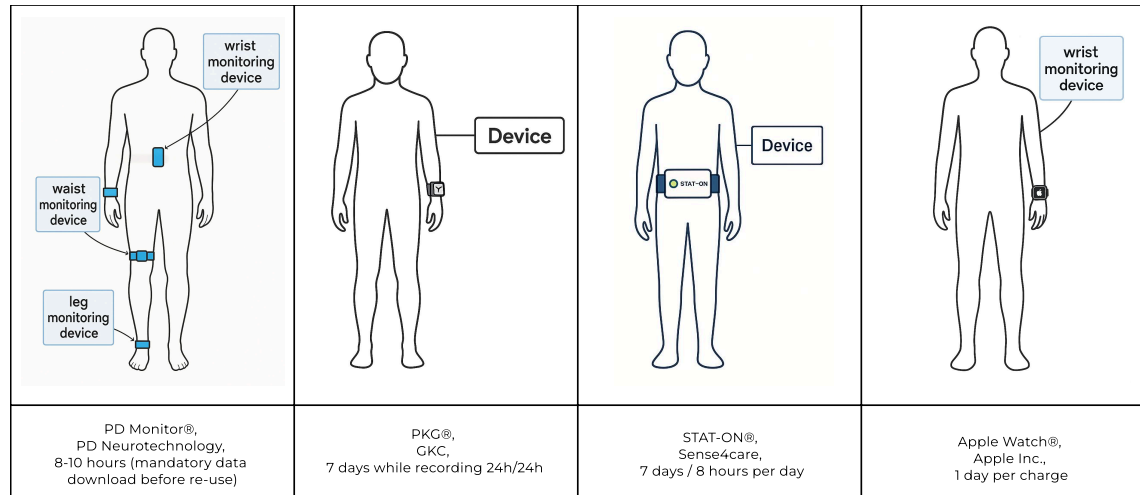


Figure 2.2: Overview of body-worn sensors for PD management, showing device placement, manufacturer, and battery life. Adapted from [6].

Several mobile applications have also been developed for symptom tracking and disease management in PD. The APDA Symptom Tracker, developed by the American Parkinson Disease Association (APDA), allows patients to manually record their symptoms and medication use, presenting the data in graphical form [16].

Parkinson's LifeKit offers daily symptom tracking using voice, movement, and cognitive assessments, while Steady Hands provides tremor analysis through drawing-based tasks [17, 18]. MyCognition focuses on cognitive health monitoring using gamified cognitive assessments, and mPower 2 collects smartphone sensor data during structured tasks for use in large-scale research studies [19, 20].

More recently, Parky, an FDA-cleared, prescription-based application, has emerged as an innovative tool for passive tracking of tremor and dyskinesia using an Apple Watch. By integrating HealthKit data, automated symptom monitoring, and medication reminders, Parky enhances communication between patients and healthcare professionals (HCP), supporting optimized treatment decisions [21].

2.4 Summary and Knowledge Gaps

There are currently several mHealth solutions for PD that show promising potential. However, most of these tools are either aimed at patient self-management or developed for use in isolated research projects. Many require high patient engagement or focus only on motor symptoms, leaving important gaps in capturing non-motor features and enabling

large-scale research. Additionally, while devices such as PDMonitor, PKG, and STAT-ON provide valuable clinical insights, their multi-sensor setups and complex reporting limit their usability in everyday settings.

This project addresses these limitations by designing a mobile application for early-to-mid-stage PD patients that integrates Apple Watch based data collection with patient-reported information. The application is specifically designed to support longitudinal research, enabling a more comprehensive understanding of both motor and non-motor symptom fluctuations over time and in daily life.

3 Design Methods

The design process of ParkTrack followed a user-centered and iterative approach, grounded in the UX lifecycle, as illustrated in Figure 3.1. This approach aimed to keep the needs, preferences, and contexts of PD patients and researchers central throughout the design process. By engaging patients, caregivers, and clinicians at multiple stages, the approach was able to integrate practical insights into the conceptual design and refine the prototype based on direct stakeholder feedback.

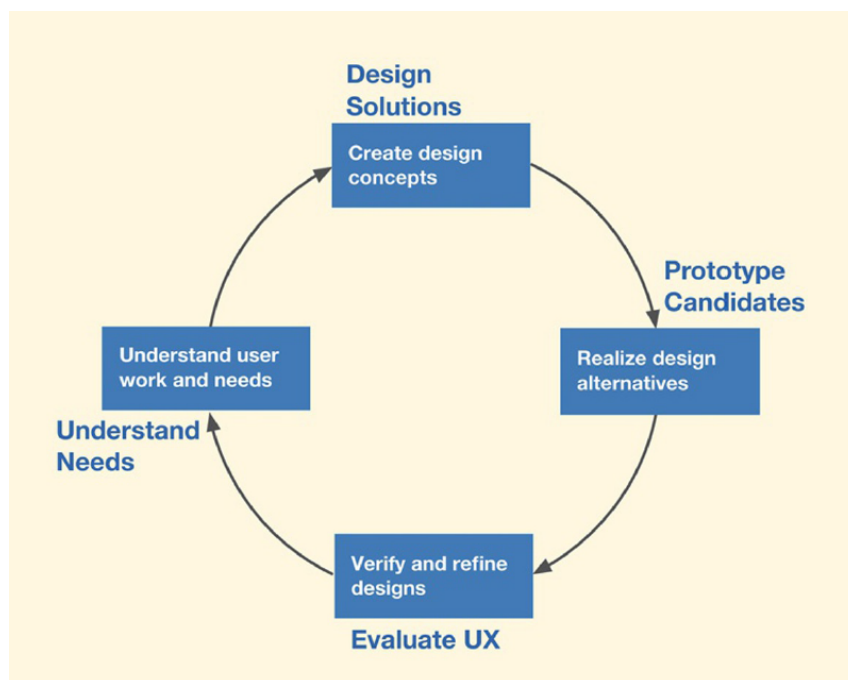


Figure 3.1: The basic UX design lifecycle process in terms of elemental lifecycle activities. Taken from the book [8]

The following sections describe the applied methods, structured according to the main stages of the UX lifecycle: (1) Understanding User Needs, (2) Creating Design Concepts, (3) Prototyping, and (4) Evaluating UX.

3.1 Understanding User Needs

The first phase focused on exploring the challenges faced by PD patients in their daily lives, as well as the information researchers require for clinical studies. To achieve this, qualitative methods were applied:

Clinical observations: A three-hour observation was carried out at Department of Neurology, Movement Disorder Clinic, Roskilde Hospital, following a neurologist during patient consultations and a nurse during patient assessments. This provided first-hand insights into clinical workflows, existing assessment practices, challenges in monitoring symptoms, and approaches to data collection.

Co-design meetings: Two structured co-design workshops were held at the Movement Disorder Clinic at Rigshospitalet, involving three patients, two relatives, a neurologist, a physiotherapist, and my clinical supervisor, who has a research background in PD. The first session aimed to gather expectations and needs related to symptom tracking, medication logging, cognitive tasks, and wearable integration. The second session focused on providing feedback on the initial prototype.

Pre-workshop preparation: Prior to the first co-design workshop, a list of reflective questions was shared with the patient and relative participants to help them prepare for the discussion and clarify their expectations. Written responses are provided in Appendix A.

The collected data were analyzed thematically to identify functional and non-functional requirements, such as ease of use for older adults, integration with wearable sensors, and meaningful data visualization. These requirements formed the foundation for subsequent design decisions.

3.2 Creating Design Concepts

Based on the identified needs, several design artifacts were created to structure and prioritize features for the ParkTrack prototype. A Lean UX Canvas was developed prior to the first workshop to articulate the problem space, user pain points, and value propositions, serving as a framework for structuring early design ideas and preparing for discussions with the advisory group at Rigshospitalet (see Appendix B).

In addition, a User Story Map was created, with the first iteration developed prior to the initial workshop. This artifact helped translate user goals into concrete functionalities and prioritize features for the prototype (see Appendix C).

Together, these conceptual tools supported the definition of a Minimum Viable Product (MVP) scope, balancing feasibility with the needs of the advisory group.

3.3 Prototyping

The conceptual designs were translated into progressively refined prototypes. Initial low-fidelity wireframes were created to outline the application's structure and navigation flow, serving as a foundation for discussion during the first co-design workshop (see Appendix E).

Building on the feedback and prioritized features from this workshop, a high-fidelity interactive prototype was subsequently designed in Figma (see Appendix F). This prototype incorporated core functionalities, including medication reminders, motor and non-motor symptom tracking, cognitive exercises, and wearable data visualization.

It was presented during the second co-design workshop, where participants had the opportunity to interact with the system and provide feedback on usability, content, and visual design.

By moving from abstract sketches to an interactive prototype, the design process enabled

iterative refinement of both content and functionality, supporting that the prototype evolved in line with user needs and project objectives.

3.4 Evaluating UX

Evaluation was embedded throughout the process using formative feedback methods. Feedback from co-design workshops with patients, relatives, and clinicians was used to identify usability issues, clarify meaningfulness, improve navigation, and optimize how data was presented.

The evaluation provided important findings, such as the need for simplified graphs, a more user-friendly way to log medication intake, and a daily summary showing how the user felt at different times of the day. These findings directly informed updates to the Figma prototype and guided recommendations for future development.

3.5 Summary of Methods

In summary, the design of ParkTrack combined clinical observations, engagement with an advisory group at Rigshospitalet, and iterative prototyping to support the prototype of a mobile application in addressing real-world needs. Guided by the UX lifecycle from The UX Book [8], the applied methods provided a structured framework for translating complex challenges in Parkinson's disease management into a digital health tool that supports research and patient self-monitoring.

4 Design Process

The design process of ParkTrack was iterative and user-centered, based on the UX life-cycle by Hartson and Pyla (Figure 3.1). Through continuous involvement of patients, caregivers, and healthcare professionals (HCP), the design evolved from initial ideas to a refined prototype. This chapter outlines the main stages: early exploration, feedback sessions, wireframing, prototyping, and reflections.

4.1 Initial Design Exploration and Idea Presentation

The initial design exploration aimed to define the core objectives and features of ParkTrack and support alignment with the needs of both patients and researchers.

To guide this process, a Lean UX Canvas was created (Appendix C), outlining assumptions about target users, their needs, and the intended value of the application prototype. This canvas served as a foundation for early discussions and helped clarify the project vision.

In addition, a User Story Map was created to visualize key activities and user interactions within the app (Appendix D). This map provided an overview of essential features such as symptom tracking, medication reminders, sleep monitoring, and cognitive tasks, forming the basis for subsequent design iterations.

To support the co-design discussions, a mind map was prepared to present features inspired by literature, existing Parkinson's apps, and insights gained from preliminary observations at Roskilde Hospital (Appendix G). This visual overview facilitated dialogue with participants at the first co-design session at Rigshospitalet, where feedback was collected on which features were most relevant and feasible for research-oriented self-monitoring.

4.2 First Feedback Session

The first co-design session provided valuable insights from the participants. These inputs were translated into concrete system requirements to guide the design (see Appendix G .1 for full transcription).

“... it's important to wear a watch ... that's why I forget to take my medication.”

– Patient 2

“... it would be better to have an option like 'Remind me later' ...”

– Physiotherapist and patient in workshop 2

Requirement 1: The system must integrate with wearable devices to provide medication reminders directly on the user's wrist.

Requirement 2: The system must allow flexible reminders, enabling users to postpone rather than dismiss notifications.

“... important to be able to see the wearing-off effect ...”

– Relative 1

“... helpful to use the data to explain any changes in behavior ...”

– Relative 2

Requirement 3: The system must visualize medication effects over time, including before-and-after states.

Requirement 4: The system must provide longitudinal data visualization to support retrospective analysis of behavioral changes.

“... a comment field ... to describe how they felt ...”

– Relative 1

“... possible to speak into a microphone ...”

– Relative 2

Requirement 5: The system must allow users to add contextual comments to their data entries.

Requirement 6: The system must support voice-to-text input for easier symptom logging.

“... gain insight into the connection between blood pressure and symptom patterns ...”

– PD nurse

Requirement 7: The system must enable the collection and analysis of blood pressure data in relation to symptom patterns.

“... good to know how much sleep you've had ... then you understand why you're tired ...”

– Patient 2

Requirement 8: The system must track and visualize sleep patterns to help interpret their impact on daily well-being.

“... text should be in large letters ... easy to read and understand.”

– Patient 3

Requirement 9: The system must follow accessibility guidelines by offering large, readable text and a clear interface.

These insights formed the foundation for refining the app’s core functionalities and guided the next iteration of wireframes and prototypes.

4.3 Wireframing and Prototyping

Based on the requirements derived from the first co-design session, an initial low-fidelity wireframe was created to outline the app's structure and core functionalities. This wireframe served as the foundation for designing a high-fidelity prototype in Figma, focusing on usability, clarity, and alignment with the identified user needs.

A complete overview of the wireframe and prototype flowcharts is provided in Appendix D and E, offering a visual representation of how the different screens and modules are interconnected. The following sections present the final high-fidelity prototype designed in this project, structured around six key functional areas, each illustrated with representative screens and an explanation of their purpose.

Onboarding

The onboarding flow introduces users to ParkTrack and guides them through essential setup steps, including granting data-sharing permissions, language selection, registration, gender, date of birth, and symptom stage selection (Requirement 1).

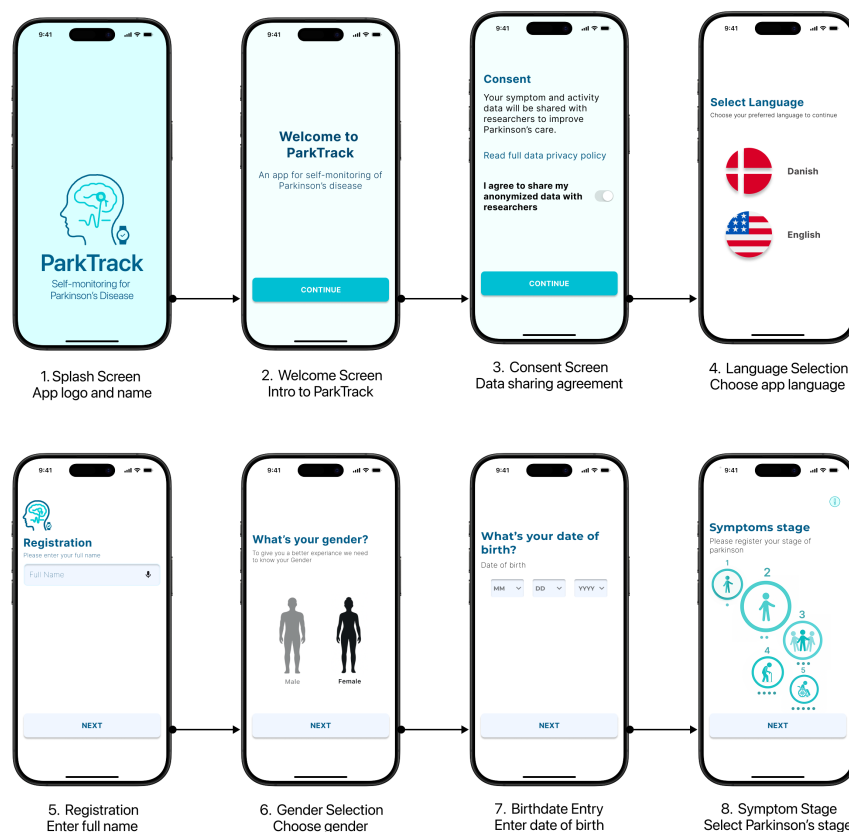


Figure 4.1: Onboarding screens of the ParkTrack prototype

Medication Tracking

The medication module allows users to log their medication intake, dosage, and timing. Notifications can be configured for reminders, with flexible scheduling options (Requirement 2). The design emphasizes simplicity and quick interaction to accommodate users with motor difficulties.

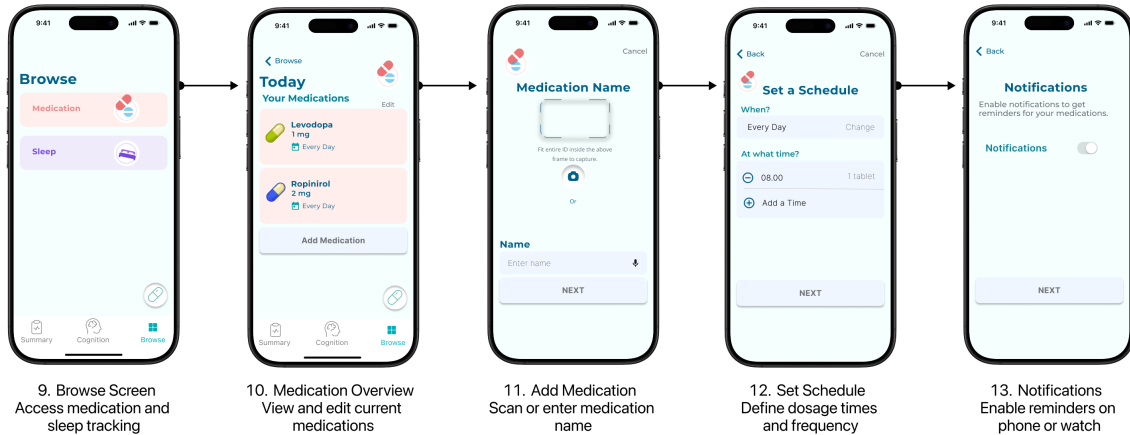


Figure 4.2: Medication tracking screens of the ParkTrack prototype

Sleep Monitoring

The sleep module integrates with the Apple Watch for automated tracking while also allowing users to manually log sleep schedules (Requirement 8). This dual approach provides flexibility for patients who may not always use wearable devices.

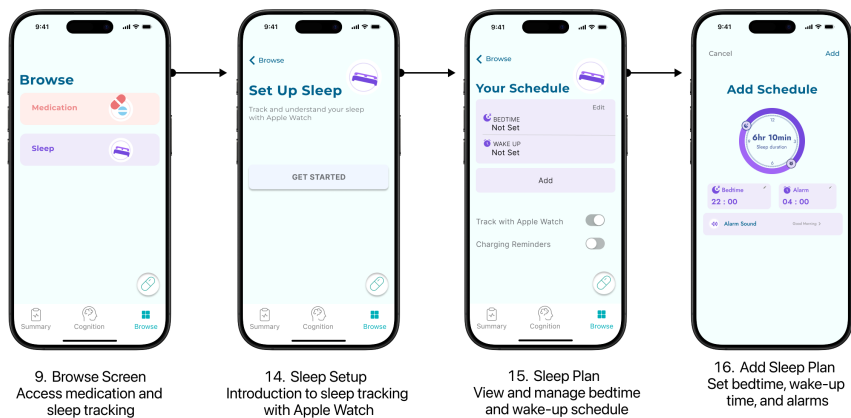


Figure 4.3: Sleep tracking screens of the ParkTrack prototype

Summary Dashboard

The summary dashboard visualizes collected data, including symptoms, medication usage, sleep patterns, and physiological measures such as pulse and blood pressure (Requirements 3, 4, 7). The visualizations help patients and researchers identify correlations and trends over time, such as medication wearing-off effects.

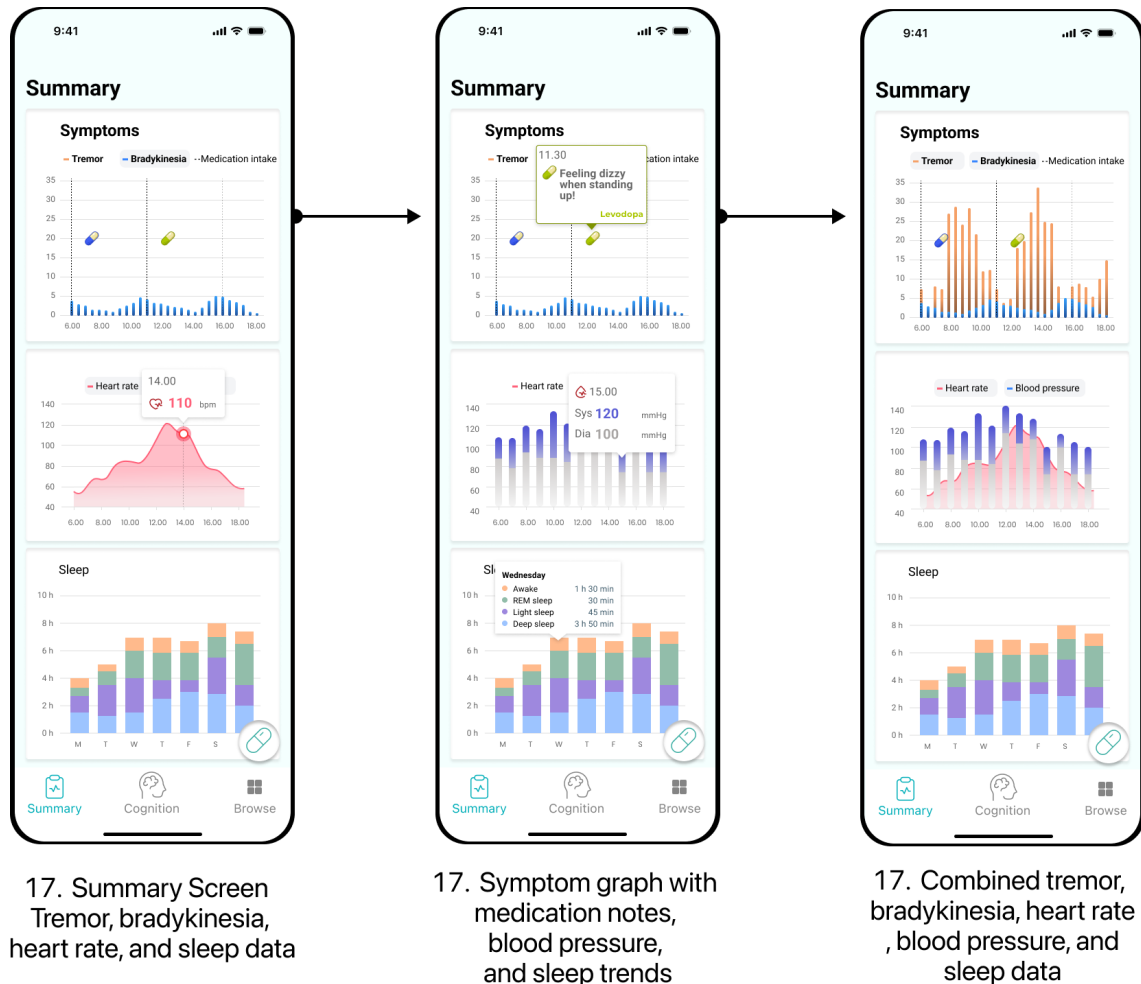


Figure 4.4: Summary dashboard of the ParkTrack prototype

Cognitive Tasks

To address Requirement 6, the prototype includes small cognitive exercises, such as the Stroop Test, that assess memory, attention, and executive functions. This module provides engaging and clinically relevant insights into cognitive performance.

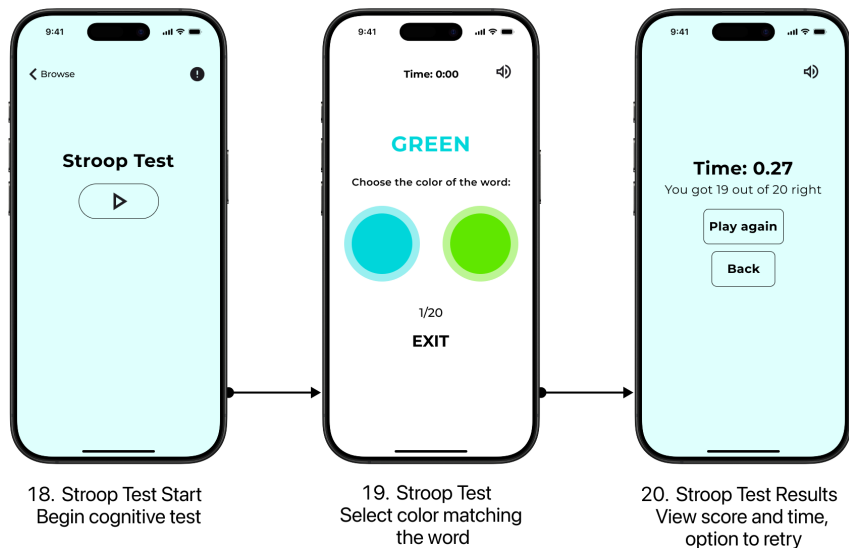


Figure 4.5: Cognitive task screens of the ParkTrack prototype

Apple Watch Integration

An Apple Watch interface was designed to facilitate passive data collection (e.g., tremor, bradykinesia, heart rate, blood pressure) and allow users to initiate measurements directly from the watch (Requirements 1, 7). This integration reduces the burden of manual input and enables continuous symptom monitoring.

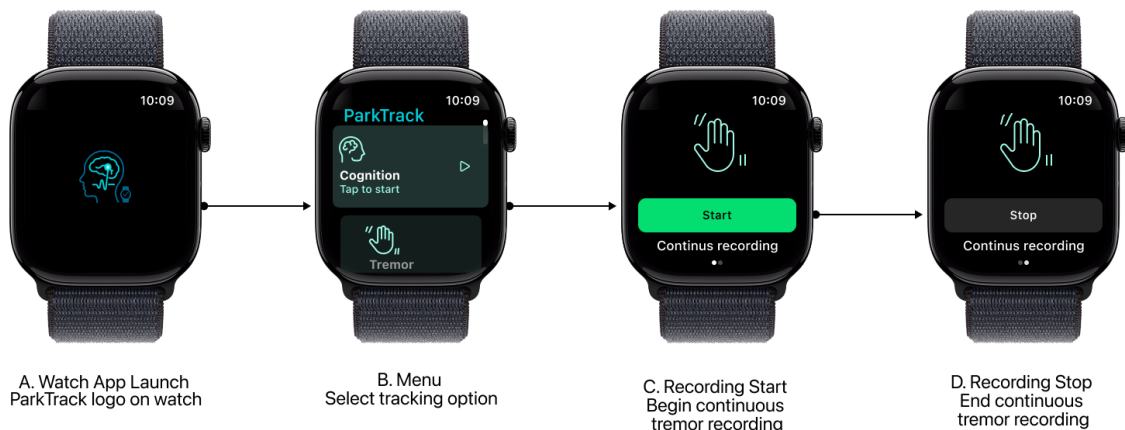


Figure 4.6: Apple Watch screens of the ParkTrack prototype.

This prototype consolidated the requirements gathered during the first iteration into a functional, user-centered design. It provided a tangible basis for the second co-design session, where participants evaluated the app's usability and suggested improvements for the next iteration.

4.4 Second Feedback Session

The second co-design meeting focused on evaluating the high-fidelity prototype and gathering feedback for further refinement. The session was held with participants including patients, relatives, and HCP, who explored the prototype on mobile. Figure 4.7 illustrates participants providing feedback on the ParkTrack prototype during this session at Rigshospitalet.



Figure 4.7: Participants providing feedback on the ParkTrack prototype during the second co-design session at Rigshospitalet.

Their insights identified several areas for improvement regarding functionality, usability, and clinical relevance. The full transcription of this meeting is available in Appendix G.2, and the full prototype is included in Appendix E.

Medication Reminders and Flexibility

“I have an issue with fixed times ... it would be much more useful if the app worked with intervals ...”

– Patient 1

“There should be an option to choose whether you want fixed times or intervals — and ideally, the possibility to use both.”

– Physiotherapist

Requirement 1: The system must allow users to set medication reminders using either fixed times, intervals, or a combination of both.

“It would be very helpful if the time of intake was shown as well.”

– Physician

Requirement 2: The system must display the actual intake time of medications alongside scheduled times to provide better tracking.

Visual Representation and Data Understanding

“On the graphs, it’s important to be able to see the wearing-off effect ...”

– Relative 1

“Heart rate and blood pressure could be summarized at the end of the day ... instead of having to check each individual timepoint.”

– PD nurse

Requirement 3: The system must visualize the wearing-off effect of medications and provide summarized daily overviews for heart rate and blood pressure values.

Onboarding and Disease Staging

“Maybe it’s better to write how many years they’ve had the disease ... even if you provide a description of the five stages, I think it would still be difficult for patients to define exactly which one they’re in.”

– PD nurse

“Otherwise, it should be more indirect — like asking whether symptoms are only on one side ... That way, the staging becomes more implicit.”

– Physician

Requirement 4: The onboarding process must simplify disease staging by allowing users to provide indirect indicators (e.g., duration since diagnosis, symptom distribution) instead of requiring self-assessment of stages.

Cognitive Testing

“Are you thinking it should all result in one combined score, or how do you imagine it?”

– PD nurse

"I think it's more about domain-specific scoring ... then you could generate an overall domain score."

– Physician

"If my mobile phone has a larger screen, I'd rather use the phone than the Apple Watch."

– Patient 2

Requirement 5: The system must include domain-specific scoring for cognitive tests and support that these tests are conducted consistently on the same device (preferably the mobile phone).

Usability and Icons

"It feels a bit confusing ... Maybe there could be a different icon to indicate a side effect, a symptom, or something else ... I just think it's something about medication."

– Patient 1

"Couldn't it just be shown as a thought bubble?"

– Relative 1

Requirement 6: The system must revise iconography to differentiate between medication, symptoms, and side effects for improved usability.

Continuous Measurements and Battery Considerations

"The way the Apple Watch works right now ... I would expect that to happen automatically."

– Physician

"I know from another project that continuous measurement takes quite a lot of battery ... But ideally, you'd want to measure everything ... Otherwise, you'd need to have all the measurements running at once."

– Physician

“Maybe you could go into the settings and choose to turn something off ... so by default, everything is enabled — but if there’s something you don’t want, you can actively disable it yourself.”

– Patient 1

Requirement 7: The system must support automatic continuous measurements (e.g., heart rate, tremor) with customizable options in settings to balance battery usage and data collection.

These insights provided concrete directions for refining the prototype and informed a set of requirements for the next design iteration, which are further discussed in Chapter 5.

5 Results and Discussion

This chapter presents the key results of the ParkTrack design process and discusses their implications. It summarizes the final prototype designed through two co-design iterations, highlights how user input shaped the solution, and reflects on the strengths and limitations of the project. The chapter concludes with recommendations for future app development.

5.1 Key Results

The final prototype (Appendix E) integrated features identified as essential by participants during the co-design sessions. These included:

Onboarding: A simplified flow for registration, indirect disease staging, and entry of date of diagnosis.

Medication tracking: Logging with flexible scheduling (fixed times or intervals) and Apple Watch integration.

Sleep monitoring: Automated data collection via Apple Watch, complemented by manual input.

Summary dashboard: Visualization of symptom progression, medication effects, sleep patterns, and physiological data (e.g., heart rate (HR), blood pressure).

Cognitive tasks: Inclusion of domain-specific tests (e.g., Stroop test) conducted on the smartphone.

Apple Watch integration: Quick-access tracking for motor symptoms and physiological measures.

Across both sessions, 16 requirements were derived (see Sections 4.2 and 4.4), capturing functional, clinical, and usability needs. These requirements demonstrate how patient, caregiver, and clinician input directly informed the design.

The design process successfully aligned the prototype with user needs, combining passive wearable data collection with options for manual input, as requested by participants. The visualizations were adjusted to make it easier for users to interpret changes in symptoms before and after medication intake, ‘wearing-off’ effects. In addition, the onboarding process was simplified, and the iconography was revised to make the interface more intuitive and accessible.

The co-design approach was essential in supporting that ParkTrack addressed both clinical and patient needs. Healthcare professionals (HCP) contributed by recommending indirect methods for disease staging, daily summaries of physiological data (e.g., minimum, maximum, and average HR and blood pressure), and cognitive tests with domain-specific

scoring. Patients emphasized usability, leading to the inclusion of larger text options, simplified navigation, and customizable settings for continuous measurements.

5.2 Limitations

Despite these achievements, the project had limitations. Only one prototype was designed, and while feedback from the second co-design session informed future changes, no additional iteration was implemented. The prototype was neither implemented nor tested in real-life settings, and the cognitive tests integrated into the design were limited in scope. These constraints limit the current validation of the solution.

5.3 Future Work

Future work should focus on implementing the refinements identified in the second co-design session (Appendix G.2), including expanded cognitive testing, enhanced data visualizations, and further personalization of reminders, giving users greater control over how and when they receive medication notifications (e.g., fixed schedules, interval-based reminders, and 'remind me later' options). Pilot testing in clinical and home environments will be essential to evaluate usability, feasibility, and data quality. Additionally, integrating more advanced analytics could enhance the tool's value for research on PD progression.

6 Conclusion

This thesis addressed a critical gap in Parkinson's disease (PD) research: the lack of comprehensive, structured, and patient-driven tools for collecting standardized, multi-modal symptom data reflecting patients' daily experiences based on a co-designed platform. Through the design of ParkTrack, a prototype of a mobile application, the project explored how mobile technologies and wearable sensors can be designed to enable reliable, self-generated data collection among early- to mid-stage PD patients, with the aim of supporting research on symptom progression and treatment outcomes.

The project employed an iterative, user-centered design (UCD) approach involving patients, relatives, and healthcare professionals (HCP) at multiple stages. This supported that the prototype of ParkTrack was both usable and acceptable for its target users. Key design elements such as simplified onboarding with entry of diagnosis date, flexible medication reminders, clear visualizations of symptom and medication patterns, domain-specific cognitive tasks, and integration with Apple Watch were directly informed by stakeholder feedback. These features enable the collection of multi-modal data, combining wearable sensor information with patient-reported outcomes, to create structured datasets for longitudinal research.

This work successfully achieved its primary goal of designing a research-oriented prototype that supports patient-driven self-monitoring of motor and non-motor symptoms, thereby addressing the needs of both patients and researchers. By minimizing manual input, providing meaningful feedback, and offering personalized features, the design supports patient engagement and adherence to self-monitoring. As such, ParkTrack demonstrates the potential of participatory design in creating tools that not only meet research needs but also empower patients to actively contribute to the understanding of their condition.

While the prototype was not tested in real-world settings and represents only a first iteration, it establishes a strong foundation for future app development and clinical evaluation. ParkTrack represents an important step toward integrating patient-driven data into Parkinson's research, empowering individuals with PD and advancing the understanding of disease progression and treatment outcomes.

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Acronyms

APDA	American Parkinson Disease Association.
API	Application Programming Interface.
H&Y	Hoehn and Yahr scale.
HCP	Healthcare Professionals.
HR	Heart Rate.
IMU	Inertial Measurement Unit.
MDS-UPDRS	Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale.
mHealth	Mobile Health.
PD	Parkinson's Disease.
PKG	Parkinson KinetiGraph.
UCD	User-Centered Design.
UX	User Experience.

A Pre-workshop preparation material

Deltagerinformation

Udfyld venligst følgende oplysninger. Disse bruges til at forstå deltagergruppens baggrund og sætte jeres feedback i kontekst.

Navn (valgfrit): _____

Rolle: ☐ Patient ☐ Pårørende ☐ Sundhedsprofessionel

Alder: _____

Køn: _____

Antal år med Parkinson (eller som pårørende): _____

Primære symptomer (kun patienter): _____

Tidligere brug af sundhedsapps: ☐ Ja ☐ Nej

Bemærk: Oplysningerne anvendes kun til projektformål og anonymiseres i rapporten.

Spørgsmål til patienter og pårørende

1. Hvad tænker I om idéen med at bruge en app til selvmonitorering af Parkinson-symptomer?

Super god idé. Hjælper mig til bedre at kunne forstå og navigere i min sygdom, hvis de monitorerede data er valide og relevante for min Parkinson. Foregår monitoreringen kontinuerligt i baggrunden (passivt) eller skal jeg aktivt "tilslutte" et registreringsdevice? Hvem skal appen kunne hjælpe/hvem er målgruppen? Kræves der særlige skills af brugeren? Er der flere målgrupper samtidigt, det vil sige både patient/pårørende og neurolog/læge? At en app "monitorerer sundhedsdata" betyder vel, at den indsamler, opbevarer og kan analysere information om min sundhed og fysiske aktiviteter. Kunne data omfatte ting som antal skridt, puls, søvn (længde og kvalitet), blodtryk, iltmætning, medicinindtag mm?

2. Hvordan foregår det i dag, når I eller en pårørende skal registrere symptomer (fx tremor, søvn, medicin)?

Jeg benytter mig af et handlingsplanssystem jeg selv har lavet, som jeg kalder for Shake. Det har jeg med, når jeg er til møde med neurologen. Så har jeg altid en opdateret status på det pågældende symptom eller gene, som jeg gerne vil tale om. Efter mødet med neurologen eller andre sundhedsprofessionelle opdaterer jeg mit skema. Så ved jeg selv og mine pårørende, hvor langt jeg er nået i udredningen af det enkelte problem eller gene.

3. Hvilke funktioner i en sådan app vil være mest værdifulde for jer?

Set fra min stol, så er det helt essentielle, medicinering, træning og pleje af sociale relationer. Vigtig at tage medicinen. Den korrekte medicin, i den rette dosis på rette tidspunkt. Overholde medicintider i forhold til fødeindtag. Huske at drikke tilstrækkeligt væske før kl. 18.00. Sikre, at jeg har en sund tarm (føde, bivirkninger andet). Ellers tænker jeg søvn, blodtryk, iltmætning, skridt, aktive timer, træning, sociale aktiviteter. Mange af disse "målinger" findes i dag i andre app's, smartwatches, stand alone devices, så hvis denne app formår at samle data ét sted og tilmed kan bearbejde dem skematisk og grafisk så ser det godt ud.

4. Hvilke udfordringer kan I forestille jer ved at bruge en app som denne?

Hvis ikke den måler/monitorerer kontinuerligt aktivt i baggrunden, så skal vi selv taste data ind. Det kan være grænseoverskridende, vanskeligt at forstå hvad man skal (manglende intuitiv betjening), små "taster" kan være svære at ramme (talegenkendelse, kan virkelig ende ud i noget volapyk). Endelig kan alle manuelle indtastninger være fyldt med slåfejl. Foretages monitoreringen automatisk kan inddata ligeledes være behæftet med store usikkerheder. Eksempelvis har jeg fået oplyst, at mit Garmin smartwatch, som jeg bruger til overvågning af min søvn, er behæftet med en usikkerhed på ca. 60%. Og hvad er målingen så værd?

5. Hvad skal der til, for at I føler jer trygge ved at bruge appen (fx ift. data, kompleksitet)?

Intuitiv funktionsbetjening, høj datasikkerhed og gennemsikuelighed af data. Hvordan monitorerer app'en? Kræves der tilslutning af diverse, måleudstyr? Hvor valide er de data der monitoreres? Hvor gemmes data? Hvordan tilgås data? Kan de tilgås fra andre enheder? Hvilke parter er involveret? Tredjepart?

6. Hvad synes I om designet og layoutet af de skærbilleder, I har set?

7. Hvilken type information vil I gerne have vist i en oversigt eller graf?

Træningshistorik, søvnmønstre og andre sundhedsvaner for at give indsigt i egen sundhed og udvikling. Monitorering af symptomer og medicinindtag, der kan bruges i dialog med neurolog og pårørende. Data fra forskellige kilder (smartphone, Apple/-Garmin Watch, andre apps og manuelle input), samlet i én visning. Valide og sammenlignelige data med mulighed for at dele med sundhedsprofessionelle og/eller pårørende og kontrollere, hvilke data der deles og hvornår.

Supplerende spørgsmål til pårørende

1. Hvilken rolle spiller du i forhold til den person med Parkinson?
2. Ville det være nyttigt for dig som pårørende at kunne få adgang til en symptomoversigt fra patientens app, hvis personen har givet tilladelse til det?
3. Hvad ville gøre det nemt og overskueligt for dig at bruge eller støtte op om brugen af appen?

Spørgsmål til sundhedspersonale

1. Hvordan vurderer I idéens relevans i forhold til nuværende forskningspraksis og patientinddragelse?
2. Hvad ser I som de største udfordringer ved at anvende selvmonitoreringsapps i forskning?
3. Hvilke typer data eller symptomer synes I er vigtigst at registrere i en app som ParkTrack (f.eks. tremor, søvn, puls, kognition, medicinindtag)?
4. Hvordan kan en app som ParkTrack bedst understøtte forskningsbehov og klinisk anvendelse?
5. Hvad tænker I om brugen af wearable sensorer til dataindsamling? Er det realistisk i praksis?
6. Hvilke metodiske eller etiske forhold bør vi være særligt opmærksomme på i videre udvikling?
7. Hvad synes I om brugergrænsefladen, som den er skitseret i wireframes?
8. Er der noget i prototypen, der fungerer særligt godt eller som bør forbedres for at gøre appen mere brugervenlig?
9. Ville det være relevant at tilføje funktioner som taleinput, automatiske påmindelser eller feedback baseret på data i fremtidige versioner af appen?

B Lean Canvas

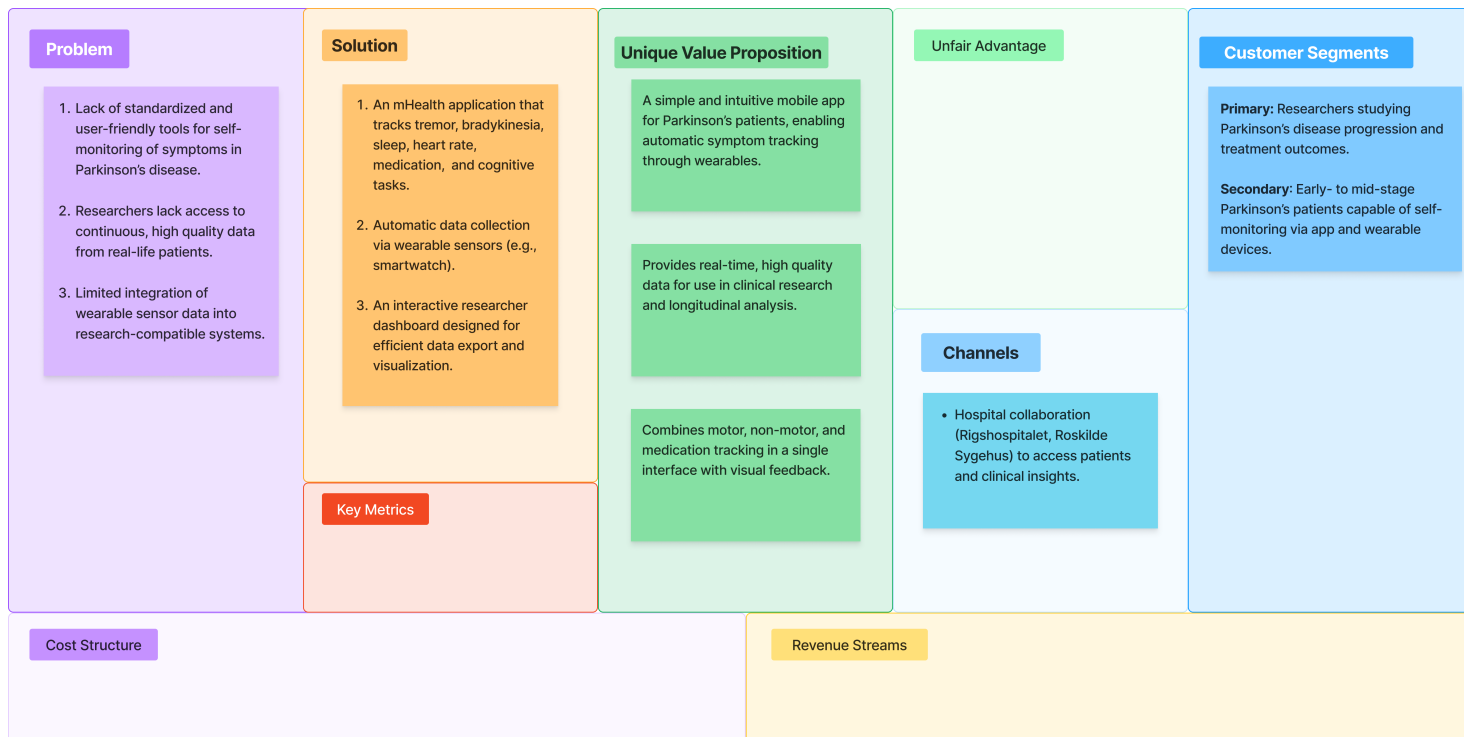


Figure B.1: Lean Canvas summarizing the problem, solution, unique value proposition, customer segments, and key considerations for the ParkTrack prototype.

C User Story Map

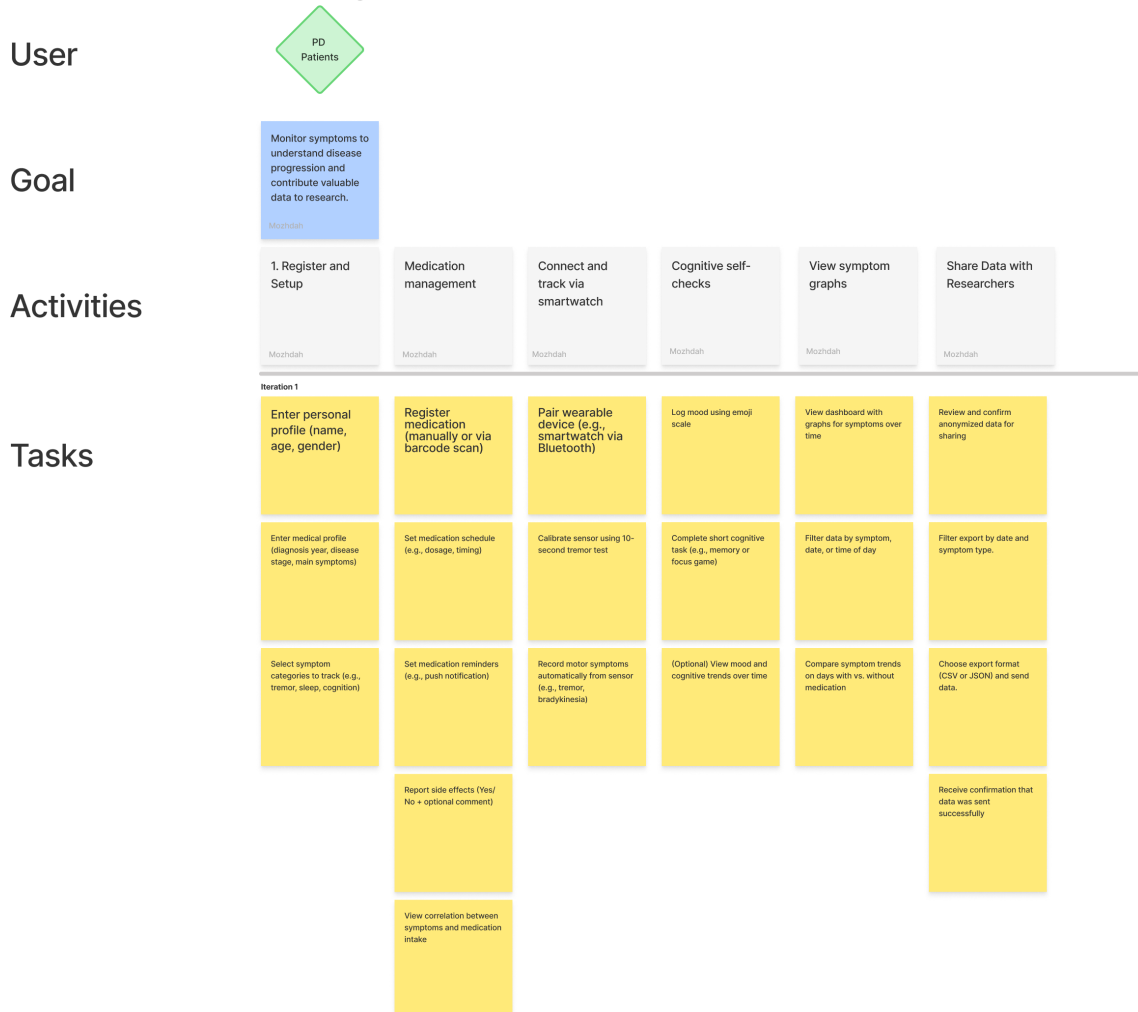


Figure C.1: User Story Map outlining core features and prioritization for the ParkTrack application based on user needs.

D Wireframe

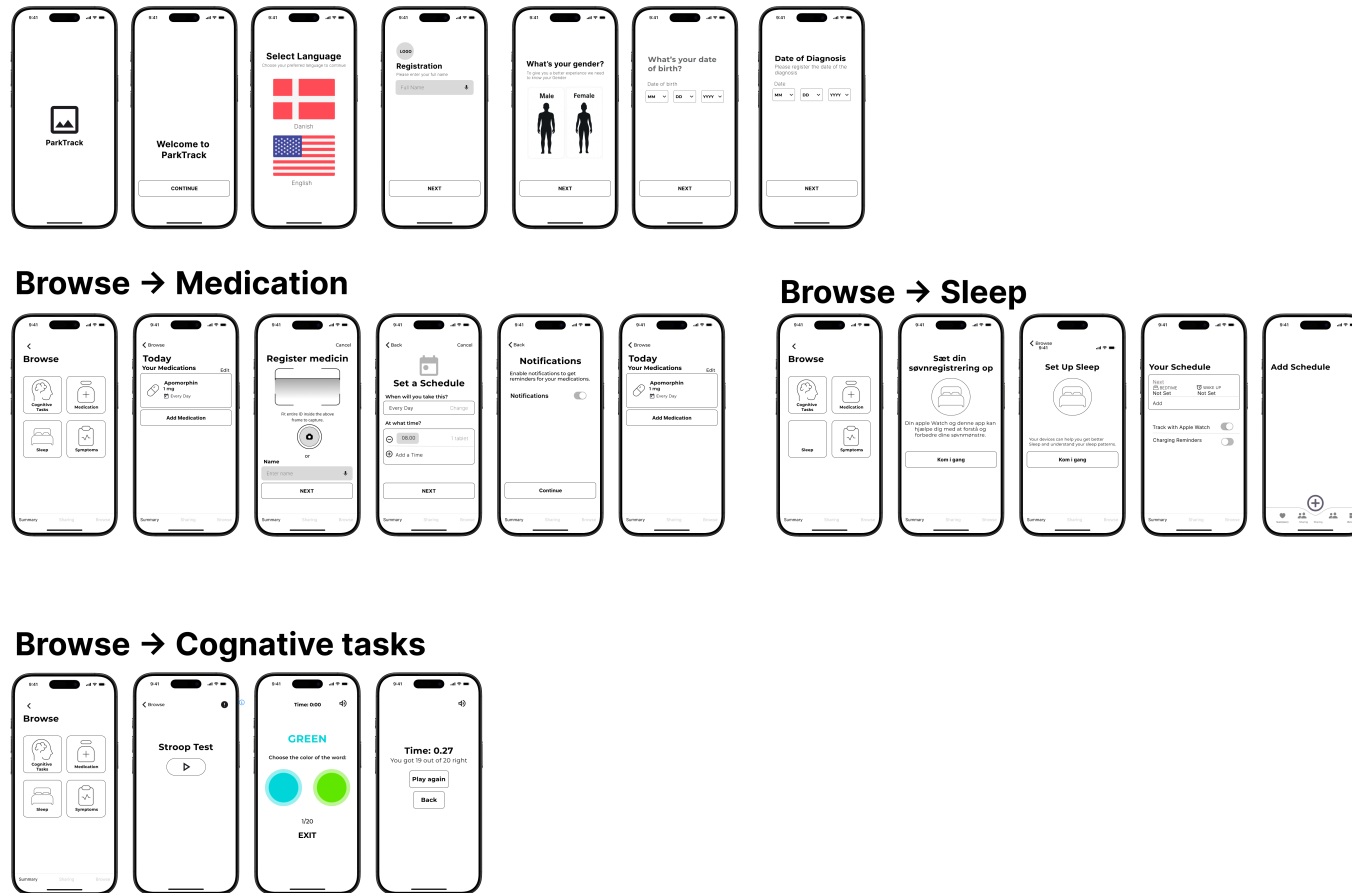


Figure D.1: Low-fidelity wireframes illustrating the initial layout and navigation flow of the ParkTrack application.

E Prototype

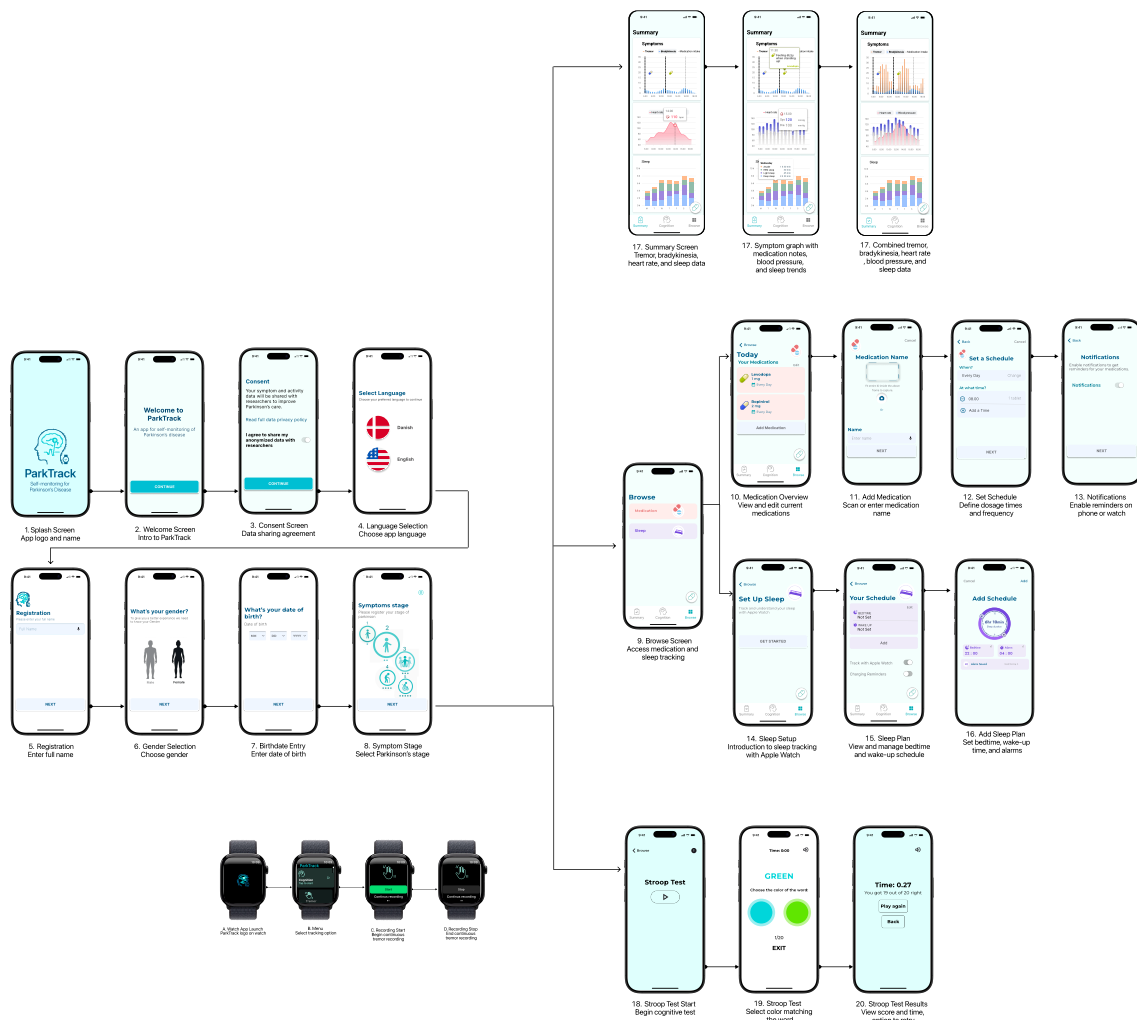


Figure E.1: High-fidelity prototype of the ParkTrack mobile application showcasing key features for self-monitoring of Parkinson's symptoms.

F Mind Map

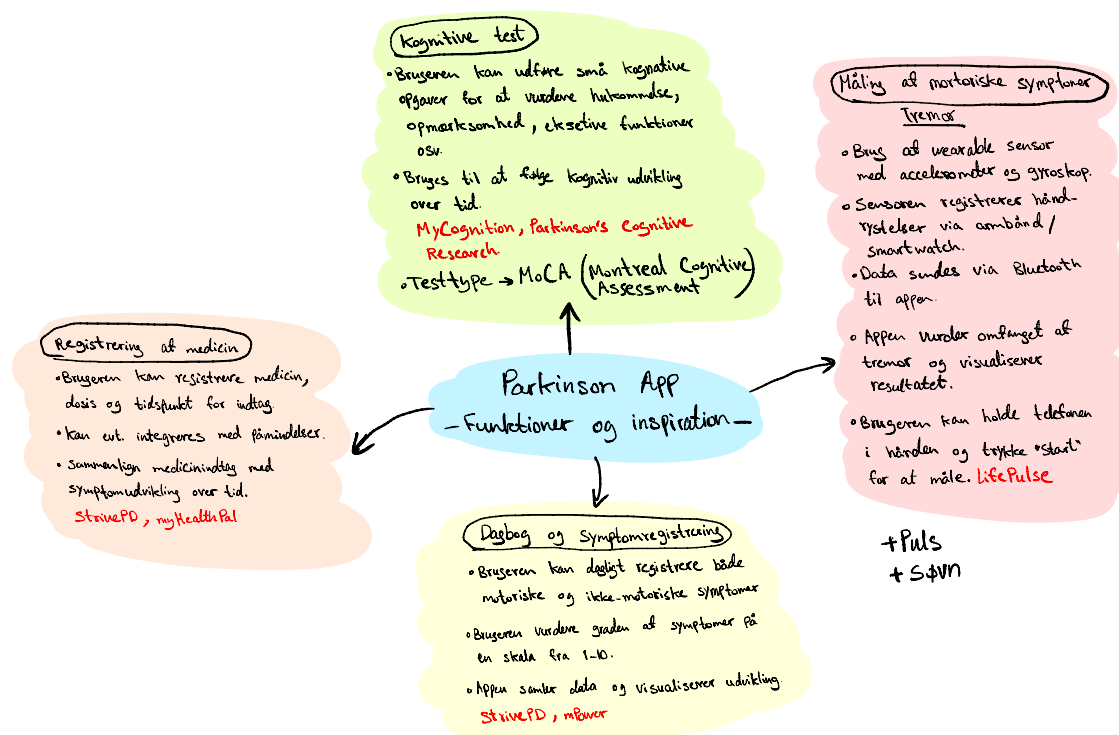


Figure F.1: Mind map illustrating the initial exploration of user needs, app features, and design considerations for the ParkTrack prototype

G Transcription of Meetings

G.1 Initial Idea Presentation and Needs Exploration

Date: June 11, 2025

Location: Rigshospitalet, Glostrup

Participants: 3 PD patients, 2 relatives, 1 physician, 1 PD nurse, 1 physiotherapist, 1 master student, and the project lead (Mozhdah).

Introduction

This first co-design meeting focused on presenting the initial concept for the prototype of a mobile application. The project lead (Mozhdah) introduced a mind map, a Lean UX Canvas, and a User Story Map to present early design ideas. These materials served as a starting point for an open dialogue with participants about their everyday challenges, preferences, and needs related to Parkinson's disease. The aim was to better understand user needs and assess whether the proposed ideas aligned with their real-life experiences.

User Quotes and Feedback

"There's a degree of uncertainty when it comes to sleep tracking especially depending on whether you use an Apple Watch or an older third- or fourth-generation device. For those who receive the data, like whether someone slept well or not, it can be hard to interpret."

– Patient 1

"With my PKG monitor, I know the level of uncertainty. I look at my data and say, okay, the uncertainty is the same every day so I might have slept more or less, but I still don't know if it was 5 or 6 hours. I've worked as a measurement technician for many years."

– Patient 1

Presented that medication can be scanned, and if scanning is not possible, the name can be entered manually. (Referenced the related user story activities.)

"It's important that there's a sound when I receive a reminder — for example, when I need to take my medication. That's really important to me."

– Patient 1

"It's important to wear a watch. I don't wear one now, and that's why I forget to take my medication."

– Patient 2

"What happens if you press 'no' when a reminder pops up?"

– Physiotherapist

"If the user presses 'no,' it's registered that the medication wasn't taken."

– Mozhdah

"If someone is busy at that moment, it would be better to have an option like 'Remind me later' instead of just saying 'no' — because medication still needs to be registered."

– Physiotherapist

"Okay, Instead of a 'no' option, we could include a 'remind me later' button."

– Mozhdah

"I'd like us to be able to see a connection between medication intake and symptoms."

– Mozhdah

"How do you envision this visually — in terms of graphs showing medication intake and symptoms over a specific period?"

– PD nurse

"On the graphs, it's important to be able to see the wearing-off effect — like how the graph looks just before taking the medication and some time after, so you can clearly see the medication's impact."

– Relative 1

"When living with Parkinson's, symptoms can vary a lot — not just from week to week or day to day, but even from hour to hour."

– PD nurse

“I definitely want to take that into account.”

– Mozhdah

Presented the feature for connecting wearable devices to the mobile app using Bluetooth.

“So you want to connect multiple devices that each measure different symptoms?”

– Relative 1

“I would like to have one device that can measure all the key symptoms included in the app like sleep, tremor, bradykinesia, heart rate, and blood pressure.”

– Mozhdah

“My biggest wish as a clinician is to gain insight into the connection between blood pressure and symptom patterns. I believe there’s much more to uncover — especially since many falls may be caused by blood pressure fluctuations.”

– PD nurse

“Some diabetes patients have implanted devices that show their blood sugar levels — maybe something similar could be used here.”

– Relative 2

“How often do you carry your phone with you?”

– Mozhdah

“Yes, we carry it with us most of the time.”

– All participants

“With a smartwatch, you can register data immediately when you receive a notification. That’s why I believe it’s best to use a smartwatch to measure symptoms — because people tend to wear it all the time.”

– Mozhdah

“I track my sleep with an Apple Watch and look at how many hours I’ve slept, how often I was awake during the night, and how much time I spent in deep sleep.”

– Patient 2

“Do you find it useful to have data about your sleep?”

– Mozhdah

“It’s good to know how much sleep you’ve had — then you understand why you’re tired. It helps others understand how you’re feeling too. I don’t experience much tremor, but I have nightmares and dream that I’m running around trying to catch someone or take control of a person.”

– Patient 2

“That’s also a symptom of Parkinson’s — patients often have very vivid dreams. So sleep is a very important parameter to include.”

– PD nurse

“Sometimes I find it difficult to see what’s on the screen. The text should be in large letters — not too small — so that it’s easy to read and understand.”

– Patient 3

Presented motor symptom registration.

“Are the measurements both quantitative and qualitative?”

– Relative 1

“It’s primarily quantitative, and the data is visualized through graphs.”

– Mozhdah

“I think it’s also important to reflect on how much the data is actually being used — for example, with PKG data. How often do you actively engage with it or make use of it in practice?”

– PD nurse

“I use it as part of tracking my personal progress. I’ve created a system for myself, where each symptom has its own number and a code indicating how severe it feels to me — on a scale from 1 to 10. I also use colors like red, yellow, and green.”

– Patient 1

“I write in my diary whenever I experience something. My diary is basically a calendar.”

– Patient 3

“What you see here on my chart, 0.4 means it’s the fourth motor symptom I experience. It’s marked in red, which means it’s the most severe and something I really feel strongly.”

– Patient 1

“In relation to my project, I’ve focused on these symptoms and would like to hear your opinion: tremor, bradykinesia, heart rate, and sleep.”

– Mozhdah

“Involuntary movements (dyskinesia) are missing — that’s also a motor symptom.”

– Relative

“When you talk about sleep, what exactly should be measured — sleep quality, disturbances? And how do you plan to record that?”

– PD nurse

“Sleep data will be collected via an external device like the Apple Watch and integrated into the app.”

– Mozhdah

"I think it works well to register data externally. It's important for me to be able to explain why I am the way I am."

– Patient 2

"What's the purpose of the cognitive feature — is it meant for training or testing?"

– Physiotherapist

"It's designed purely for testing — to give an indication of how the user is doing cognitively at a given time."

– Mozhdah

"But why couldn't it be turned into a training feature instead?"

– PD nurse

"That's definitely possible, but I think it would make the project too complex at this stage to include a training feature as well."

– Mozhdah

"If the app included some training modules — something like brain fitness or simple computer games aimed at cognitive functions, like switching strategies or maintaining an overview — do you think that's something you would use?"

– PD nurse

"I don't normally play games, but if it's an app that's relevant to my Parkinson's, I can definitely see it as a training opportunity."

– Patient 1

"I think those kinds of tasks can help train cognitive functions. But if I ask the neuropsychologists, none of them will recommend anything specific to me."

– Physiotherapist

"I use crossword puzzles — that also helps train the brain."

– Patient 2

“I just think that if it’s only used as a test, then it’s important to switch it up once in a while. Otherwise, you end up becoming quite an expert at it.”

– PD nurse

Discussed willingness to share data with researchers.

“What would the system requirements be for this app? For example, if someone wants to install it on their smartphone — what version of Android would be needed?”

– Relative 1

“It’s important to consider the users — many might have older phones, like an iPhone 10. If you build an app that requires iOS 17, which is only available on iPhone 15, you’ll lose a lot of users. Parkinson’s patients are often older and may not have the newest devices. That takes the app into a different category — one where you risk losing many users simply because the technical requirements are too high.”

– Master student

“I was thinking — maybe you could make it possible to speak into a microphone so the app converts it to text. That way, instead of writing how I feel, I could just say it. It would make it much easier for many people to record how their day has been.”

– Relative 2

“Do you think it would be helpful to have access to patient symptom data if they choose to share it with you?”

– Mozhdah

“Yeah, of course.”

– All participants

"I think it could be helpful to use the data to explain any changes in behavior. Sometimes it's easier to visualize and point out that something is getting worse, especially if the person doesn't agree. As a relative, you're often close to the person and may not notice gradual changes, but others like children or grandchildren might. Then you could go back and look at the data to see if something has changed. I really believe it could be useful."

– Relative 2

"When working with graphs, it's helpful to have detailed algorithms behind them—but what I would personally use is something like a daily overview: how I feel in the morning, before noon, in the afternoon, and in the evening. It would be great if I could simply tick off these periods, rather than seeing data hour by hour."

– Physiotherapist

"If the goal is to make the data useful for a neurologist, then it needs to be something that can be discussed at a higher level—like how the patient is doing during different parts of the day—rather than pointing to a dot on a graph at 4 p.m."

– Physiotherapist

"So it should be more qualitative. You still need the underlying data, but when visualizing, it could be in a checklist format. That's just how I imagine it being useful when visualizing graphs."

– Physiotherapist

"Could there be a comment field that allows the user to describe how they felt during a specific period? For example, after taking medication, they could add a comment about the effect two hours later—only in cases where something feels off. The idea would be to document deviations."

– Relative 1

"Yes, absolutely. That's definitely something we can include."

– Mozhdah

G.2 Prototype Feedback

Date: July 8, 2025

Location: Rigshospitalet, Glostrup

Participants: 3 PD patients, 2 relatives, 1 physician, 1 PD nurse, 1 physiotherapist, and the project lead (Mozhdah).

Introduction

This second co-design meeting focused on gathering feedback on the prototype of the mobile application. The project lead (Mozhdah) presented key features and visual concepts, including medication registration, wearable integration, and symptom tracking. Participants were encouraged to share their perspectives, experiences, and suggestions to refine the design and aim for alignment with real-world needs.

User Quotes and Feedback

Presented the onboarding screens, including data sharing permissions, language selection, name registration, gender selection, date of birth, and symptom stage selection — as shown in Screen B (see Figure 4.1).

“As a patient, do you typically know which stage of the disease you’re in?”

– Mozhdah

“Well, I don’t really know what stage I’m at. No.”

– Patient 2

“Maybe it’s better to write how many years they’ve had the disease. I know that doesn’t necessarily reflect the stage, but even if you provide a description of the five stages, I think it would still be difficult for patients to define exactly which one they’re in.”

– PD nurse

“No, I think you’re right. I believe it’s often the relatives who might be the ones placing the person into a stage, because we see it differently than the person who actually has Parkinson’s.”

– Relative 1

“Otherwise, it should be more indirect – like asking whether symptoms are only on one side or both sides, or if there are balance issues. That way, the staging becomes more implicit. I think that’s actually easier.”

– Physician

“It also depends a bit on how well the medication works. If the medication is effective, then you feel significantly better than if it doesn’t work.”

– Patient 3

“Has the clinic here ever provided an answer about which stage the patient is in?”

– Relative 2

“It’s become a formal requirement due to data collection for the national quality database. But really, it’s just something we do as part of standard reporting – it’s based on the old staging model. Whether it’s actually meaningful is up for debate. It’s more of a registration formality.”

– Physician

Presented the summary screen, which displays graphs of symptoms and their correlations — as shown in Screen B (see Figure 4.1).

“It feels a bit confusing – like, there’s an arrow and then the icon for side effects. Maybe there could be a different icon to indicate a side effect, a symptom, or something else. Because when I first see it, I just think it’s something about medication.”

– Patient 1

“It’s confusing to me that the pill icon represents something different from the arrow. I don’t have a specific suggestion right now, but when I see a pill, I immediately think it’s about taking medication.”

– Patient 1

“Couldn’t it just be shown as a thought bubble?”

– Relative 1

Explained what the pill icon represents and that the colors indicate a specific medication — as shown in Screen B (see Figure 4.1).

“Yes, but many people take multiple pills at the same time, so they won’t know which pill the arrow refers to.”

– Relative 1

“I think it’s difficult for patients to say exactly which medication is causing it, but they can usually report that they’re experiencing side effects — just not which drug specifically.”

– Physician

“It would be quite interesting to view certain parameters in relation to medication intake.”

– PD nurse

“Heart rate and blood pressure could be summarized at the end of the day. For example, it could show your lowest and highest values of the day — instead of having to check each individual timepoint.”

– Nurse and researcher

“It would also be helpful to get an overview of heart rate trends from day to day. That way, next time you visit your GP, you can say, ‘this is the range my blood pressure typically stays in,’ instead of borrowing a device to measure it three times a day for several days and writing it all down manually. You could just open the app and show a summary.”

– PD nurse

Presented the browse screen and explained how users can register their medication — as shown in Screen B (see Figure 4.1).

“It would be very helpful if the time of intake was shown as well.”

– Physician

“I have an issue with fixed times. I don’t always get up at the same time every day, so it doesn’t make sense for me to take my medication at the exact same time either. It would be much more useful if the app worked with intervals — like, if I start at 7:00, then remind me again after 4 or 5 hours. I’ve given up on fixed times because my days are just too different.”

– Patient 1

“There should be an option to choose whether you want fixed times or intervals — and ideally, the possibility to use both.”

– Physiotherapist

“It would just be great if that option were available, because in the other tools I’ve tried, it hasn’t been.”

– Patient 1

“It would be smart if you could set the first time, and then if you get up earlier or later one day, the following intervals adjust accordingly.”

– Relative 2

“If we were to show this in the app, how do you think it should be displayed?”

– Mozhdah

“I think you should be able to choose between two models: either a fixed schedule where you take your medication at the same time every day, or a rolling schedule that adjusts based on when your day starts. Before entering the medication settings, the user should select whether they want a fixed or a dynamic schedule — maybe we could call it that, ‘dynamic.’ For example, if someone wants to take their medication every five hours starting from when they wake up, they should choose the dynamic model. But if they prefer taking it at 7:00 every day, then they’d select the fixed-time model. The two options should be clearly separated — I don’t think you can really mix them. People should simply be able to choose the model that works best for them, because people have different routines.”

– Patient 1

“We do what the doctors say, right. You’re supposed to take your medication at 7:30. But you might be sleeping really well at that time — so the other model would actually be really smart.”

– Patient 2

“Does it have to be an Apple Watch?”

– Patient 1

“Yes, I chose to use the Apple Watch for this prototype because it offers reliable sensor data and integrates well with iPhones, which many patients already use. It also supports features like heart rate and motion tracking, which are relevant for Parkinson’s monitoring.”

– Mozhdah

Presented how sleep tracking is set up in the app — as shown in Screen B (see Figure 4.1).

“I don’t have an Apple Watch — I’m completely against Apple. My watch is an older model, but it can still tell when I fall asleep, based on changes in my heart rate. It can detect when I start sleeping because my heart rate changes before I fall asleep.”

– Patient 1

“So I imagine it’s more like a notification saying ‘now it’s time to sleep,’ but that doesn’t mean you actually fall asleep then. My watch can detect when I’m not really asleep yet.”

– Patient 1

“Apple works in a way where you get a notification saying it’s time to go to bed. And then, after about 15 minutes, it registers that you’ve fallen asleep.”

– Patient 2

“Well, mine definitely doesn’t work like that — and I’m actually quite happy with that.”

– Patient 1

“Is it something the user sets manually — like when they go to bed — or is it something that’s automatically measured?”

– PD nurse

“You set your own bedtime manually.”

– Patient 2

“Mine also tells me to go to bed, but it can tell that I’m not actually sleeping. When I check the app, it shows that I haven’t settled down yet, right.”

– Patient 1

Presented the cognitive testing feature in the app, specifically explaining the Stroop test — as shown in Screen B (see Figure 4.1).

“Are you thinking it should all result in one combined score, or how do you imagine it?”

– PD nurse

“I know the tests are completely different, but you could have an individual score for each, and then one that applies specifically to the instrument integrated into the app. That way, you could compare over time. I know it wouldn’t be validated, but it’s still something you could build on or refine further.”

– PD nurse

“Yes, exactly. I was also thinking that if I look more closely into the different tests, and if each one focuses on a specific cognitive area, then I could group them accordingly — for example, this one focuses on memory.”

– Mozhdah

“I think it’s more about domain-specific scoring — like what kind of score you can create within a particular domain. If you have several cognitive tests that, like the Stroop test, target specific subdomains, then you could generate an overall domain score, for example for short-term memory.”

– Physician

“The test could be further designed by adding multiple levels with varying degrees of difficulty.”

– Mozhdah

Explained the use of the Apple Watch in the app concept, including how it collects data such as heart rate, movement, and sleep patterns, and can also be used to perform cognitive tests — as shown in Screen B (see Figure 4.1).

“Should it be called ‘Tremor’?”

– PD nurse

“Yes, maybe. But I think you should think broadly. We here probably know a bit more than most people. I would use plain Danish and say ‘rysten’ shaking, because if you want both younger and much older people to try it — especially the older ones — that word might be easier to understand.”

– Patient 3

“Isn’t the screen too small on the watch to run those kinds of cognitive games?”

– Physician

“I’ve taken that into account by choosing games that don’t require much screen space — for example, the Stroop test only uses two buttons.”

– Mozhdah

“If you’re offering multiple tests, wouldn’t it be a bit odd if only some of them can be done on the watch? I don’t know, but I think it would make more sense if cognitive tests were something you always did on the phone — so users know, ‘this is where I take the test.’”

– Physician

“If my mobile phone has a larger screen, I’d rather use the phone than the Apple Watch.”

– Patient 2

Explained that the Apple Watch interface includes four symptom categories, and that pressing the start button initiates the measurement automatically — as shown in Screen B (see Figure 4.1).

“What can you really gain from measuring at a self-chosen time? I mean, then it wouldn’t be able to show a curve over the whole day, for example.”

– Relative 2

“The idea is, of course, that the measurement runs continuously. I included the manual start option in case the watch runs out of battery — this way, the user can easily restart the measurement by pressing the start button.”

– Mozhdah

“The way the Apple Watch works right now — you don’t have to manually tell it to start measuring heart rate, right? At least when it comes to heart rate, I would expect that to happen automatically.”

– Physician

“Blood pressure can be entered manually.”

– PD nurse

“I know from another project that continuous measurement takes quite a lot of battery. If the watch is used all the time for this, it drains more power. But ideally, you’d want to measure everything — especially if the goal is to see continuous patterns across different time points. Otherwise, you’d need to have all the measurements running at once.”

– Physician

“Instead, maybe you could go into the settings and choose to turn something off. So by default, everything is enabled — but if there’s something you don’t want, you can actively disable it yourself.”

– Patient 1

“What do you think would be the best way to set up the measurements the first time?”

– Mozhdah

“Just start everything at once by pressing a single button.”

– *Physician*

“I think it was very logical — well done!”

– *Relative 1*

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